

Exp: 4 Design a lexical Analyzer to validate operators to recognize the operators +,-,*,/ using regular arithmetic operators using C

```
#include <stdio.h>

#include <string.h>

int main() {

    char s[5];

    printf("Enter any operator: ");

    if (fgets(s, sizeof(s), stdin) != NULL) {

        s[strcspn(s, "\n")] = '\0';

        switch(s[0]) {

            case '>':

                if(s[1] == '=') printf("Greater than or equal\n");

                else printf("Greater than\n");

                break;

            case '<':

                if(s[1] == '=') printf("Less than or equal\n");

                else printf("Less than\n");

                break;

            case '=':

                if(s[1] == '=') printf("Equal to\n");

                else printf("Assignment\n");

                break;

            case '!':

                if(s[1] == '=') printf("Not Equal\n");

                else printf("Bit Not\n");

                break;

            case '&':

                if(s[1] == '&') printf("Logical AND\n");

                else printf("Bitwise AND\n");

                break;

        }

    }

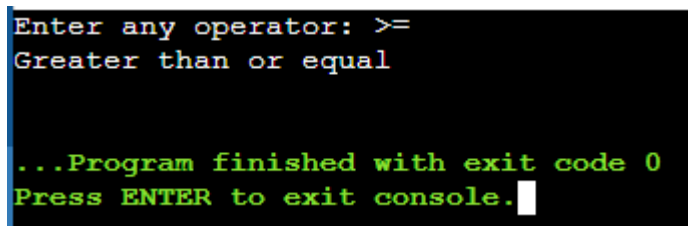
}
```

```

    case '|':
        if(s[1] == '|') printf("Logical OR\n");
        else printf("Bitwise OR\n");
        break;
    case '+': printf("Addition\n"); break;
    case '-': printf("Subtraction\n"); break;
    case '*': printf("Multiplication\n"); break;
    case '/': printf("Division\n"); break;
    case '%': printf("Modulus\n"); break;
    default: printf("Not an operator\n");
}
}
return 0;
}

```

Output:



```

Enter any operator: >=
Greater than or equal

...Program finished with exit code 0
Press ENTER to exit console.

```